

**ROCKFIELDS SENIOR SCHOOL –ICT DEPARTMENT
ROBOTICS & BASIC ELECTRONICS FINDINGS REPORT**

DATE: 23.11.2025

A. Robotics Kits (A group of 3-5 students)

Item	Estimated Price (KES)
Arduino Uno R3	KES 1,600
HC-SR04 Ultrasonic Sensor	KES 200
SG90 Servo Motor	KES 300
L293D Motor Driver	KES 350
2-Wheel Robot Chassis Kit	KES 1,250
Jumper Wires Set(3 different types)	KES 150
Breadboard	KES 200
Battery Holder (4xAA)	KES 150
Component Storage Box	KES 200
LCD IC2	KES 750
Bluetooth Module	KES 800
DHT 11	KES 300
1 and 4 Digit 7 Segment	KES 50 & 200
Buzzer	KES 50
Remote sensor	KES 200
Sensor Kits	KES 1800
Joy Stick	KES 350

B. Basic Electronics Teaching Kit

Item	Estimated Price (KES)
LED Pack (20 pcs)	KES 150

Assorted Resistors Pack (100 pcs)	KES 200
Soldering Iron	KES 450
Potentiometers (10 pcs)	KES 300
3.3V Battery rechargeable + Charger	KES 400 + 250
Buzzer	KES 100
Bulb 5v	KES 50
Push Buttons (10 pcs)	KES 150
1 & 2 way on off switches	KES 30
Multimeter (Digital)	KES 1,200

Recommended Nairobi Suppliers

KTechnics – Arduino, sensors, robotics
 Nerokas Kenya – Sensors, motor drivers, electronics
 Novatech Electronics – Robotics parts
 Pixelelectric – Motors, robotics kits
 Sagan EElectronics – general electronics

To determine the number of Components needed it will depend on the number of student present i.e. **We have 100 students when we divide that by 5 we will get 20 groups so we will buy 20 pieces in each component.**

Alternatively we can buy as a full kits but each kit will be shared per group (**I kit cost 7,500sh**)

TRAINING STRUCTURE

General Details for Such Trainings (Typical Structure)

- **Venue:** Rockfield's Junior School
- **Target Audience:** Teachers (Primary to Senior High School levels) with little to no prior experience in coding and robotics.

- **Training Focus:**
 - **Introduction to Coding:** Using block-based platforms like **Scratch** or entry-level Python.
 - **Hands-on Robotics:** Building and programming simple robots using kits like **LEGO® SPIKE™ Prime, LEGO® MINDSTORMS®, Arduino, or Micro: bit.**
 - **Curriculum Integration:** How to incorporate these skills into existing subjects like Science, Mathematics, and Technology.
 - **Pedagogy:** Effective teaching methods for STEM subjects.

Estimated Costs (Can Vary Widely)

The cost is highly dependent on the duration, the equipment provided, and the organizing body.

- **Short Workshop (6 Days):** Could range from **KES 30,000 for teachers below 20** and **KES 40,000 for above 21 teachers.**
- **Comprehensive Course (e.g., 2 weeks):** Will cost **KES 50,000 for unlimited teachers.**
- **What's often included:** Training materials, certificate of participation. Robotics kits are usually provided for use *during* the training but not always to take home.

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